

Exam Summary

Module Name

Student

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Week 1: Linear Algebra for Deep Learning

Chapter 1: Linear Algebra for Deep Learning

Part 1 — The Building Blocks

Type	Dims	Notation	Example
Scalar	0D	a	single loss value
Vector	1D	$\mathbf{x}$ — always column	$[1, 2, 3]^\top$
Matrix	2D	$\mathbf{A}$, element $A_{i,j}$	weight layer
Tensor	3D+	$\mathbf{A}$, element $A_{i,j,k}$	image batch

Column vector and matrix notation:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \quad \mathbf{A} = \begin{pmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{2,2} \\ A_{3,1} & A_{3,2} \end{pmatrix}$$

Part 2 — Operations

Matrix addition (same size only):

- Used for: combining two sets of features, adding residuals in ResNets

$$C_{i,j} = A_{i,j} + B_{i,j}$$

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 6 & 8 \\ 10 & 12 \end{pmatrix}$$

Scalar on matrix (a = scale, c = bias/shift):

- Used for: normalising data, applying a weight + bias to a whole layer

$$D_{i,j} = a \cdot B_{i,j} + c$$

$$2 \cdot \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + 10 = \begin{pmatrix} 12 & 14 \\ 16 & 18 \end{pmatrix} \rightarrow \text{every element } \times 2, \text{ then } +10$$

Broadcasting (vector added to every column of matrix):

- Used for: adding a bias vector to an entire batch of inputs at once

$$C_{i,j} = A_{i,j} + b_j$$

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} + \begin{pmatrix} 10 \\ 20 \end{pmatrix} = \begin{pmatrix} 11 & 22 \\ 13 & 24 \\ 15 & 26 \end{pmatrix}$$

Matrix multiplication:

- *Used for:* the core forward pass — transforming inputs through a weight layer: $\mathbf{y} = W\mathbf{x}$

$$\mathbf{C} = \mathbf{AB} \quad C_{i,j} = \sum_k A_{i,k} \cdot B_{k,j}$$

Shape: $(m \times k) \cdot (k \times n) = (m \times n)$ — inner dims must match.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}$$

Inner / dot product (vectors only, result is a scalar):

- *Used for:* measuring similarity between two vectors; attention scores in Transformers

$$\mathbf{x}^\top \mathbf{y} = \sum_i x_i y_i = \|\mathbf{x}\|_2 \|\mathbf{y}\|_2 \cos \theta$$

$$(1 \ 2 \ 3) \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} = 1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6 = 32$$

Hadamard (element-wise) product:

- *Used for:* gating mechanisms (LSTM, GRU), applying masks, dropout

$$\mathbf{C} = \mathbf{A} \odot \mathbf{B} \quad C_{i,j} = A_{i,j} \cdot B_{i,j}$$

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \odot \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 5 & 12 \\ 21 & 32 \end{pmatrix}$$

Properties of matrix multiplication:

- Distributive: $A(B + C) = AB + AC$
- Associative: $A(BC) = (AB)C$

Exam Trap: $AB \neq BA$ — matrix multiplication is **NOT commutative**. This is on every exam.

Transpose:

- *Used for:* converting column↔row vectors, computing dot products ($\mathbf{x}^\top \mathbf{y}$), flipping weight matrices in backpropagation

$$(A^\top)_{i,j} = A_{j,i} \quad (AB)^\top = B^\top A^\top$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}^\top = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix} \rightarrow (2 \text{ times } 3) \text{ becomes } (3 \text{ times } 2)$$

Part 3 — Norms

- *Used for:* measuring magnitude of weights, gradients, errors; regularisation (L1/L2); comparing vectors

General L^p norm:

$$\|x\|_p = \left(\sum_i |x_i|^p \right)^{\frac{1}{p}}, \quad p \geq 1$$

Example with $x = [3, -4]^\top$:

$$\|x\|_1 = 3 + 4 = 7 \quad \|x\|_2 = \sqrt{9 + 16} = 5 \quad \|x\|_\infty = 4$$

Norm	Formula	Use
L^2 (Euclidean)	$\ x\ _2 = \sqrt{\sum_i x_i^2}$	most common
Squared L^2	$\ x\ _2^2 = x^\top x$	easier to compute
L^1	$\ x\ _1 = \sum_i x_i $	sparsity
L^0	count of non-zeros	not a true norm
L^∞	$\ x\ _\infty = \max_i x_i $	max element
Frobenius	$\ A\ _F = \sqrt{\sum_{i,j} A_{i,j}^2} = \sqrt{\text{tr}(AA^\top)}$	matrix size

Quick Reference — A vs $A^\top$ vs A^{-1}

Notation	Name	What it does	How it looks
A	Original matrix	Transforms input	unchanged
$A^\top$	Transpose	Swaps rows and columns	$(A^\top)_{i,j} = A_{j,i}$
A^{-1}	Inverse	Undoes A — like dividing	$A^{-1}A = I$

Starting with the same matrix A :

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

Transpose $A^\top$ — flip across diagonal (rows become columns):

$$A^\top = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} \rightarrow \text{row 1 } [1, 2] \text{ becomes col 1 } \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

Inverse A^{-1} — the “undo” matrix, found by calculation (not just flipping!):

$$A^{-1} = \begin{pmatrix} -2 & 1 \\ 3/2 & -1/2 \end{pmatrix} \rightarrow A^{-1}A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

Special case — Orthogonal matrix: transpose = inverse ($A^\top = A^{-1}$)
 $\rightarrow$ only works because orthogonal matrices purely rotate (no stretching).

Exam Trap: $A^\top$ is always easy — just flip.

A^{-1} requires real computation — unless A is orthogonal (then $A^{-1} = A^\top$ for free).

Part 4 — Identity, Inverse, Solving Systems

- Used for: solving systems of equations $Ax = b$; finding model parameters given data

$$A^{-1}A = I \quad Ax = b \Rightarrow x = A^{-1}b$$

Example — solve $Ax = b$:

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix} \quad b = \begin{pmatrix} 6 \\ 8 \end{pmatrix} \rightarrow A^{-1} = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/4 \end{pmatrix} \rightarrow x = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

How to compute A^{-1} — 2×2 formula (exam):

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \quad \text{where } \det(A) = ad - bc$$

Steps: ① swap a and d on diagonal ② flip signs of b and c ③ divide by $\det(A)$

Example:

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \rightarrow \det(A) = 1 \cdot 4 - 2 \cdot 3 = -2 \rightarrow A^{-1} = \frac{1}{-2} \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 3/2 & -1/2 \end{pmatrix}$$

$$\text{Verify: } A^{-1}A = \begin{pmatrix} -2 & 1 \\ 3/2 & -1/2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \checkmark$$

Exam Trap: If $\det(A) = ad - bc = 0 \rightarrow$ division by zero $\rightarrow$ **no inverse exists** (singular matrix).

For larger matrices: use Gauss-Jordan — write $[A \mid I]$, row-reduce until left side $= I$, right side becomes A^{-1} .

A^{-1} exists only if: A is square AND columns are linearly independent (non-singular).

Situation	Solutions
$m > n$ (overdetermined)	None (generically)
$m < n$ (underdetermined)	Infinitely many
$m = n$, independent cols	Exactly one

Part 5 — Special Matrices

Diagonal:

- Used for: efficient scaling; eigenvalue matrices in decompositions; fast inversion

$$\text{diag}(v)x = v \odot x \quad \text{diag}(v)^{-1} = \text{diag}([1/v_1, \dots, 1/v_n]^T)$$

$$\text{diag}\left(\begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}\right)x = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2x_1 \\ 3x_2 \\ 4x_3 \end{pmatrix} \rightarrow \text{same as } \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} \odot x$$

Symmetric:

- Used for: covariance matrices, distance matrices, any matrix of pairwise values

$$A = A^T$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 4 \\ 3 & 4 & 6 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 4 \\ 3 & 4 & 6 \end{pmatrix}^{\top} \rightarrow \text{mirrored across diagonal}$$

Orthogonal (rows AND cols are orthonormal):

- *Used for:* rotation matrices, V and U in SVD, efficient inversion (just transpose)

Exam Trap: Symmetric $\neq$ Orthogonal — completely different properties!

Symmetric: $A = A^{\top}$ (mirrored across diagonal).

Orthogonal: $A^{\top}A = I$ (columns perpendicular + unit length). A matrix can be orthogonal without looking symmetric at all.

How to check if a matrix is orthogonal — verify two conditions on every column:

1. **Unit length:** $\|c_i\|_2 = 1$ — each column has length 1
2. **Perpendicular:** $c_i^{\top}c_j = 0$ for all $i \neq j$ — every pair of columns is at 90°

If both hold $\rightarrow$ orthogonal $\rightarrow$ use $A^{-1} = A^{\top}$ (free inverse, no computation).

Example — check col 1 and col 2 of $A = \frac{1}{3} \begin{pmatrix} 2 & -2 & 1 \\ 1 & 2 & 2 \\ 2 & 1 & -2 \end{pmatrix}$:

First, extract the columns (each column = one vertical slice of the matrix):

$$c_1 = \frac{1}{3} \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} \quad c_2 = \frac{1}{3} \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix} \quad c_3 = \frac{1}{3} \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$$

Check unit length — square each element, sum them, take the square root. Must equal 1:

$$\|c_1\|_2 = \sqrt{\left(\frac{2}{3}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2} = \sqrt{\frac{4}{9} + \frac{1}{9} + \frac{4}{9}} = \sqrt{\frac{9}{9}} = \sqrt{1} = 1 \checkmark$$

Check perpendicular — dot product = multiply matching elements, then sum. Must equal 0:

$$\begin{aligned} c_1^{\top}c_2 &= \frac{2}{3} \cdot \frac{-2}{3} + \frac{1}{3} \cdot \frac{2}{3} + \frac{2}{3} \cdot \frac{1}{3} \\ &= \frac{-4}{9} + \frac{2}{9} + \frac{2}{9} = \frac{-4+2+2}{9} = \frac{0}{9} = 0 \checkmark \end{aligned}$$

Same check for every other pair: $c_1^{\top}c_3 = 0$ and $c_2^{\top}c_3 = 0$ (verify yourself!).

$$A^{\top}A = AA^{\top} = I \Rightarrow A^{-1} = A^{\top}$$

Inverting an orthogonal matrix = just transposing it — no computation needed.

Part 6 — Eigendecomposition

- *Used for:* understanding what a matrix does geometrically; PCA; finding principal directions of data; quadratic optimisation

$$Av = \lambda v$$

- v = eigenvector (direction unchanged by A , unit length $\|v\| = 1$)

- λ = eigenvalue (scaling factor along that direction)

How to compute eigenvalues — solve the characteristic equation $\det(A - \lambda I) = 0$:

Step 1: Subtract λ from every diagonal element $\rightarrow$ get $A - \lambda I$

Step 2: Compute $\det(A - \lambda I) = 0 \rightarrow$ gives a polynomial in λ

Step 3: Solve for λ using the quadratic formula (for 2×2):

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Example — find eigenvalues of $B = \begin{pmatrix} 3 & 12 \\ 12 & 56 \end{pmatrix}$:

$$\text{Step 1: } B - \lambda I = \begin{pmatrix} 3 - \lambda & 12 \\ 12 & 56 - \lambda \end{pmatrix}$$

$$\text{Step 2: } \det = (3 - \lambda)(56 - \lambda) - 12 \cdot 12 = \lambda^2 - 59\lambda + 24 = 0$$

$$\text{Step 3: } \lambda = \frac{59 \pm \sqrt{59^2 - 4 \cdot 24}}{2} = \frac{59 \pm \sqrt{3385}}{2} \rightarrow \lambda_1 \approx 58.59, \quad \lambda_2 \approx 0.41$$

Example — $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$ has eigenvalues $\lambda_1 = 4, \lambda_2 = 2$:

$$A \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \end{pmatrix} = 4 \begin{pmatrix} 1 \\ 1 \end{pmatrix} \rightarrow \text{direction } [1, 1]^\top \text{ scaled by } 4$$

$$A \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} \rightarrow \text{direction } [1, -1]^\top \text{ scaled by } 2$$

Full decomposition:

$$A = V \text{diag}(\boldsymbol{\lambda}) V^{-1} \quad V = [\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(n)}]$$

Intuition: **rotate** (by V) $\rightarrow$ **scale** (by λ) $\rightarrow$ **rotate back** (V^{-1}).

For real symmetric matrices: n real eigenvalues, orthogonal eigenvectors $\rightarrow V^{-1} = V^\top$.

Optimization (exam!):

$$\max_{\|\mathbf{x}\|=1} \mathbf{x}^\top A \mathbf{x} \rightarrow \mathbf{x} = \text{eigenvector of max eigenvalue}$$

$$\min_{\|\mathbf{x}\|=1} \mathbf{x}^\top A \mathbf{x} \rightarrow \mathbf{x} = \text{eigenvector of min eigenvalue}$$

Part 7 — Singular Value Decomposition (SVD)

- *Used for:* pseudoinverse, data compression, dimensionality reduction (PCA), recommender systems, solving any linear system

Works for **any** matrix (generalization of eigendecomposition):

$$A = U D V^\top$$

- U : orthogonal — left singular vectors (output directions)

- D : diagonal — singular values (how much each direction is stretched)
- V : orthogonal — right singular vectors (input directions)

How to build D — take the square roots of the eigenvalues of $A^\top A$, put them on the diagonal in descending order. Shape of D = same shape as A :

$$\lambda_1 \geq \lambda_2 \geq \dots \rightarrow D = \begin{pmatrix} \sqrt{\lambda_1} & 0 & \dots \\ 0 & \sqrt{\lambda_2} & \dots \\ \vdots & \vdots & \ddots \end{pmatrix} \text{ same shape as } A$$

Extra rows or columns $\rightarrow$ filled with zeros.

Example — A is 3×2 , eigenvalues $\lambda_1 = 58.59$, $\lambda_2 = 0.41$:

$$D = \begin{pmatrix} \sqrt{58.59} & 0 \\ 0 & \sqrt{0.41} \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 7.65 & 0 \\ 0 & 0.64 \\ 0 & 0 \end{pmatrix} \quad (3 \times 2, \text{ same as } A)$$

How to build V — eigenvectors of $A^\top A$, stacked as columns, in same order as eigenvalues:

$$V = [\mathbf{v}^{(1)} \mid \mathbf{v}^{(2)} \mid \dots] \rightarrow \text{col 1} = \text{eigenvector of } \lambda_1, \text{ col 2} = \text{eigenvector of } \lambda_2, \dots$$

Example — eigenvalues $\lambda_1 = 58.59$, $\lambda_2 = 0.41$ give eigenvectors $\mathbf{v}^{(1)}$ and $\mathbf{v}^{(2)}$:

$$V = \begin{pmatrix} -0.211 & -0.977 \\ -0.977 & 0.211 \end{pmatrix} \quad (2 \times 2 \text{ since } A \text{ has 2 columns})$$

How to build U — eigenvectors of $AA^\top$, stacked as columns, same eigenvalue order. Shape = $m \times m$ where m = rows of A :

$$U = [\mathbf{u}^{(1)} \mid \mathbf{u}^{(2)} \mid \dots \mid \mathbf{u}^{(m)}]$$

Example — A is 3×2 so $AA^\top$ is $3 \times 3 \rightarrow U$ is 3×3 . Eigenvalues: $\lambda_1 = 58.59$, $\lambda_2 = 0.41$, $\lambda_3 = 0$:

$$U = \begin{pmatrix} -0.283 & -0.868 & 0.408 \\ -0.538 & -0.208 & -0.816 \\ -0.794 & 0.451 & 0.408 \end{pmatrix}$$

Tip: Summary — how to get each piece of SVD $A = UDV^\top$:

- V : eigenvectors of $A^\top A$ as columns (size: $n \times n$)
- D : $\sqrt{\lambda_i}$ on diagonal, same shape as A (size: $m \times n$)
- U : eigenvectors of $AA^\top$ as columns (size: $m \times m$)

Always in descending eigenvalue order.

For a general matrix the singular values in D tell you how much the matrix stretches space in each direction — largest first.

Part 8 — Moore-Penrose Pseudoinverse

- *Used for:* least-squares regression, solving overdetermined systems (more equations than unknowns); any time a true inverse doesn't exist

When A^{-1} doesn't exist:

$$A^+ = VD^+U^\top \quad (D^+ : \text{reciprocal of each non-zero diagonal of } D)$$

Example — $D = \begin{pmatrix} 3 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix} \rightarrow D^+ = \begin{pmatrix} 1/3 & 0 & 0 \\ 0 & 1/2 & 0 \end{pmatrix}$

Case	Result
More rows than cols (overdetermined)	Minimises $\ A\mathbf{x} - \mathbf{b}\ _2$ (least squares)
More cols than rows (underdetermined)	Minimises $\ \mathbf{x}\ _2$ among all solutions

Part 9 — Trace

- Used for: computing Frobenius norm, simplifying matrix derivative expressions, checking eigenvalue sums

$$\text{tr}(A) = \sum_i A_{i,i} = \sum_i \lambda_i$$

Example:

$$\text{tr}\left(\begin{pmatrix} 1 & 9 & 7 \\ 2 & 5 & 3 \\ 4 & 6 & 8 \end{pmatrix}\right) = 1 + 5 + 8 = 14$$

- $\text{tr}(A^\top) = \text{tr}(A)$
- $\text{tr}(ABC) = \text{tr}(CAB) = \text{tr}(BCA) \leftarrow$ cyclic invariant
- $\|A\|_F = \sqrt{\text{tr}(AA^\top)}$

Part 10 — Determinant

- Used for: checking if a matrix is invertible; measuring volume scaling; appears in probability density formulas (multivariate Gaussian)

$$\det(A) = \prod_i \lambda_i$$

Example:

$$\det\left(\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}\right) = 3 \cdot 3 - 1 \cdot 1 = 8 \rightarrow \lambda_1 \cdot \lambda_2 = 4 \cdot 2 = 8 \checkmark$$

- $\det(A) = 0 \rightarrow$ singular, not invertible (at least one eigenvalue is zero)
- $\det(A) \neq 0 \rightarrow$ invertible
- Geometric meaning: factor by which A scales volume

Cheat Sheet

WEEK 1 — LINEAR ALGEBRA CHEAT SHEET

Structures: scalar=0D · vector=1D(col) · matrix=2D · tensor=3D+

Multiply:

- Standard: $C_{i,j} = \sum_k A_{i,k} B_{k,j}$
- Dot product: $\mathbf{x}^\top \mathbf{y} = \|\mathbf{x}\| \|\mathbf{y}\| \cos \theta$

- Hadamard $\odot$: $C_{i,j} = A_{i,j}B_{i,j}$
- $AB \neq BA$ — NEVER commutative

Norms: $L^2 = \sqrt{\sum x^2}$ · $L^1 = \sum |x|$ · $L^\infty = \max |x|$ · $\|A\|_F = \sqrt{\text{tr}(AA^\top)}$

Special: Symmetric $A = A^\top$ · Orthogonal $A^{-1} = A^\top$ · Diagonal: invert = reciprocal diagonal

Eigen: $Av = \lambda v \rightarrow A = V \text{diag}(\lambda) V^{-1}$

SVD: $A = UDV^\top$ (always works)

Pseudoinverse: $A^+ = VD^+U^\top$

Trace: $\text{tr}(A) = \sum A_{i,i} = \sum \lambda_i$ (cyclic invariant)

Det: $\det(A) = \prod \lambda_i$ · zero = singular = not invertible